

3.2.6 F: Further vectors

	Content
F1	Understand and use the vector and Cartesian forms of an equation of a straight line in 3D.

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F2	Understand and use the vector and Cartesian forms of the equation of a plane.

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F3	Calculate the scalar product and use it to calculate the angle between two lines, to express the equation of a plane, and to calculate the angle between two planes and the angle between a line and a plane.

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F4	Check whether vectors are perpendicular by using the scalar product.

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F5	Calculate and understand the properties of the vector product. Understand and use the equation of a straight line in the form $(\mathbf{r} - \mathbf{a}) \times \mathbf{b} = 0$. Use vector products to find the area of a triangle.

F6	Find the intersection of two lines. Find the intersection of a line and a plane. Calculate the perpendicular distance between two lines, from a point to a line and from a point to a plane.
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